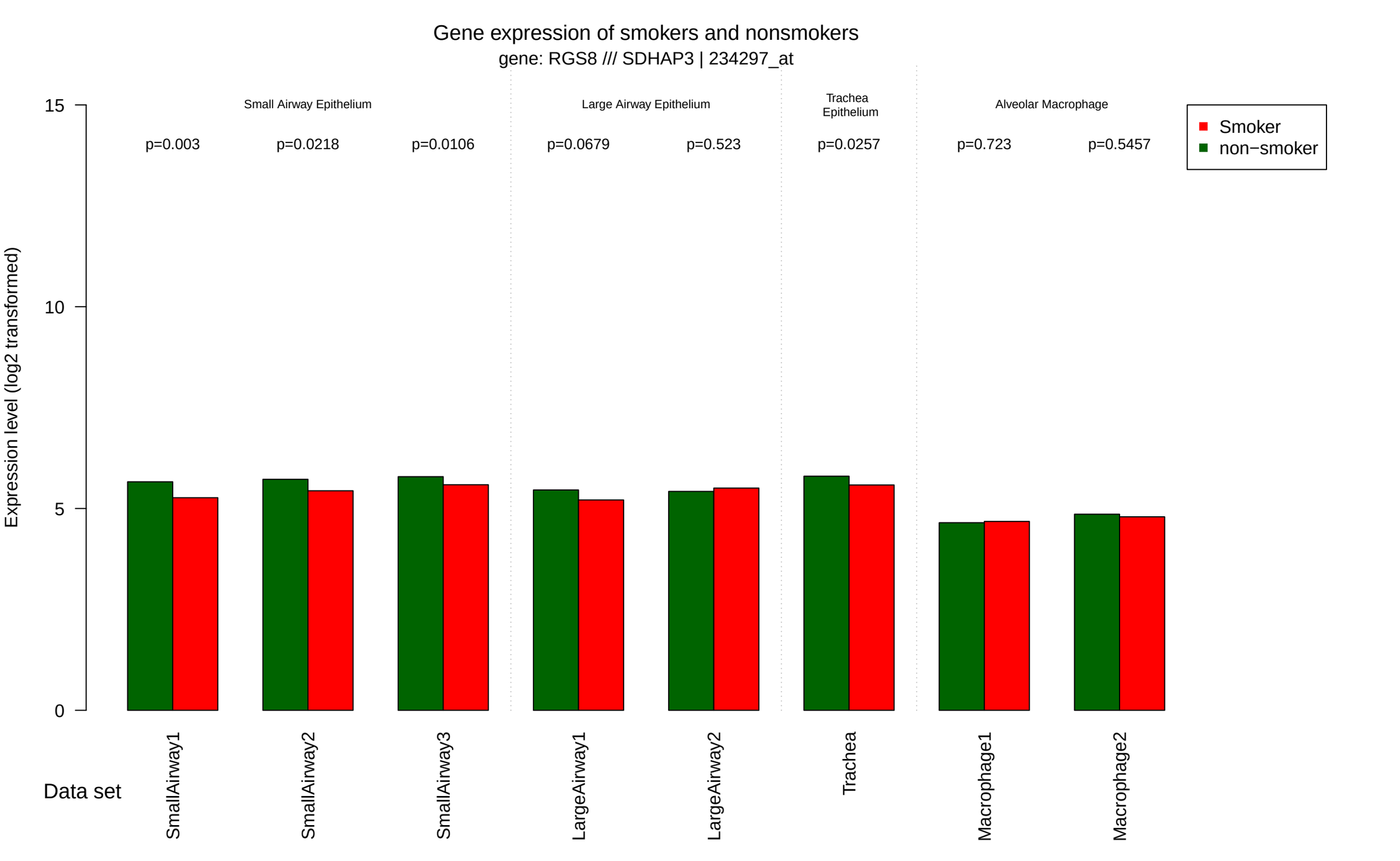
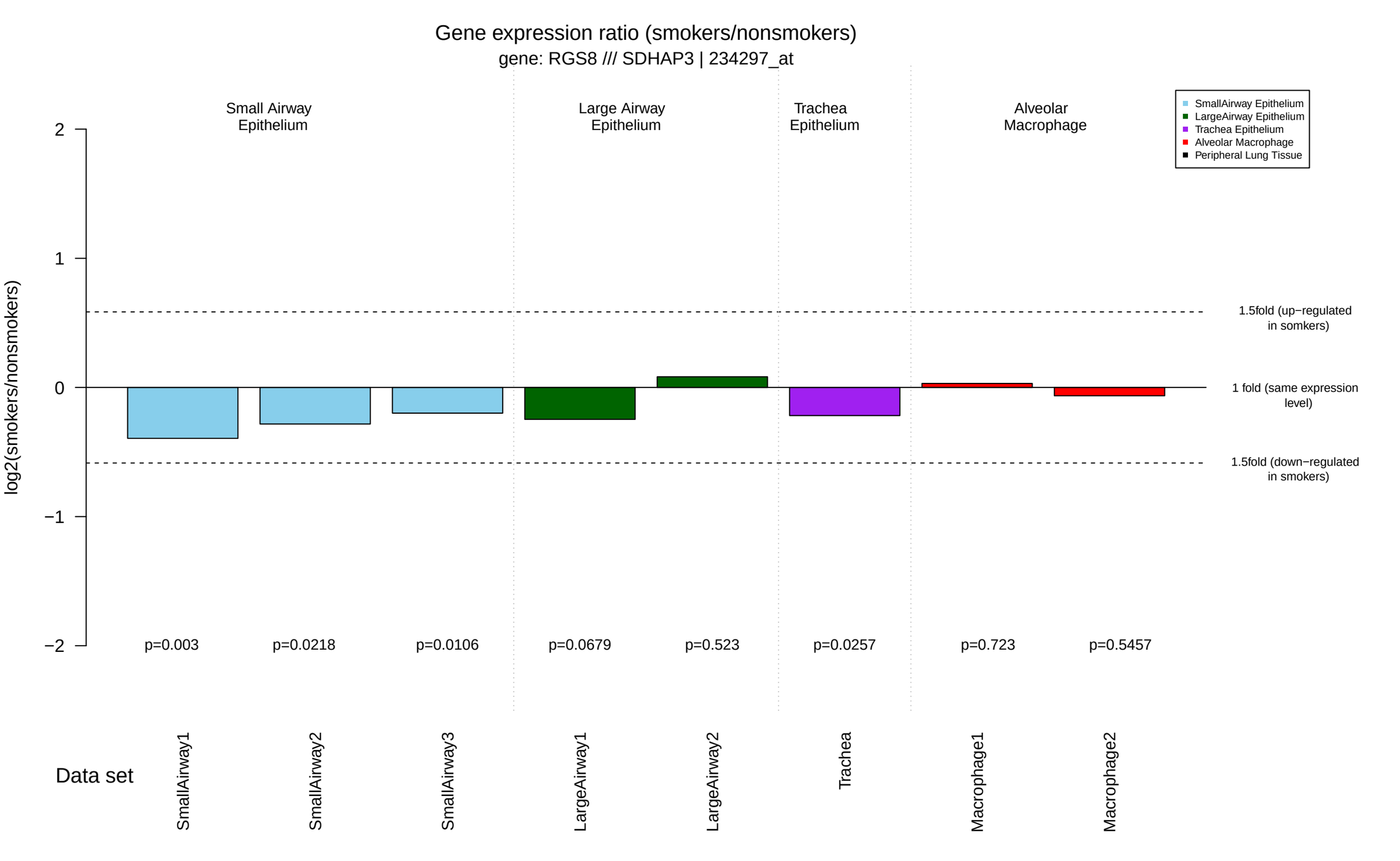
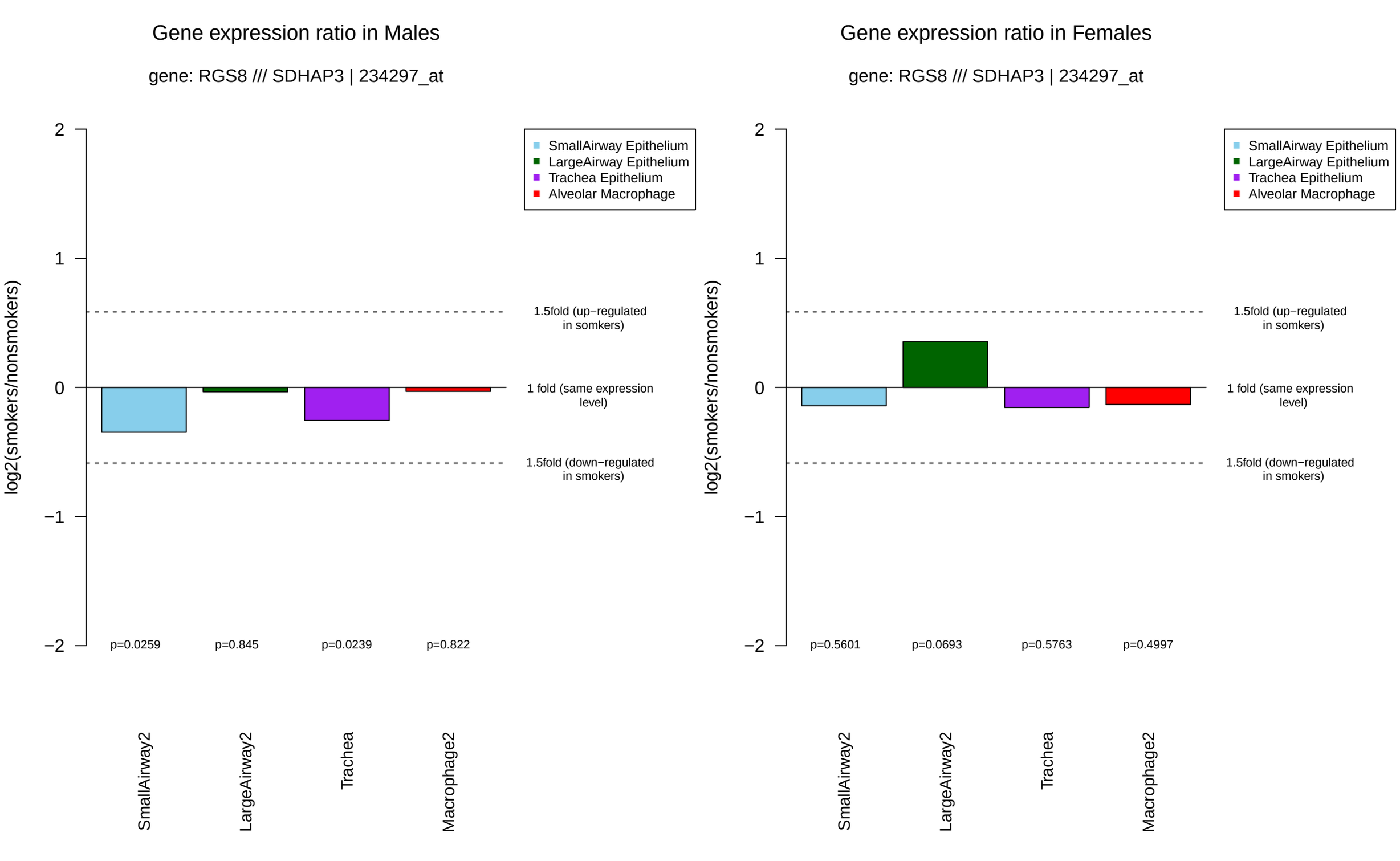
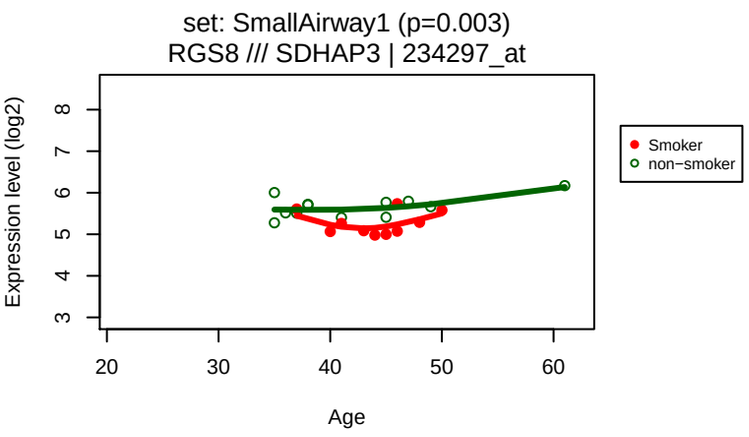
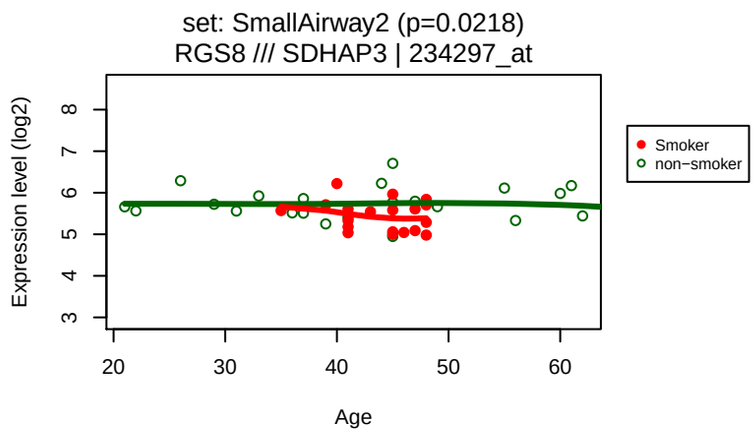
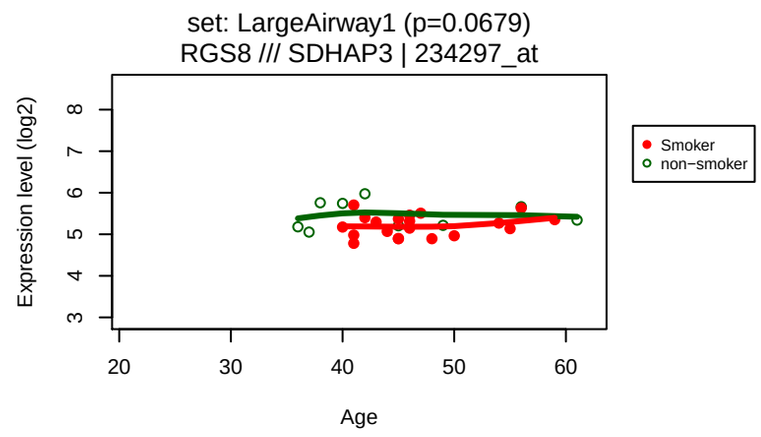

Gene expression of smokers and nonsmokers
gene: RGS8 /// SDHAP3 | 234297_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.003
p=0.0218
p=0.0106
p=0.0679
p=0.523
p=0.0257
p=0.723
p=0.5457
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

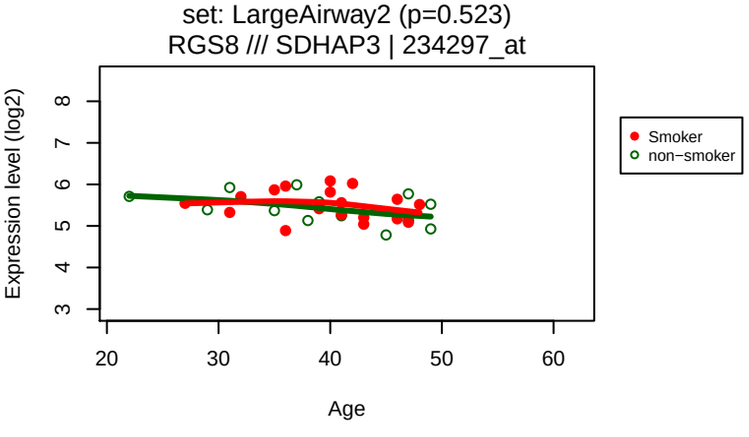

Gene expression ratio (smokers/nonsmokers)
gene: RGS8 /// SDHAP3 | 234297_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.003
p=0.0218
p=0.0106
p=0.0679
p=0.523
p=0.0257
p=0.723
p=0.5457
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

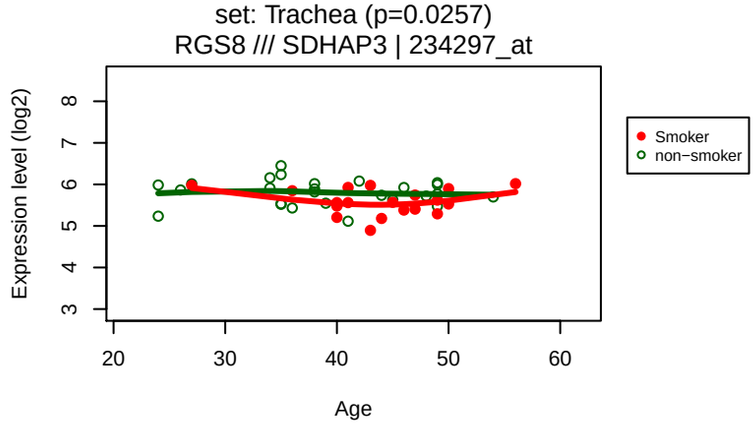

Gene expression ratio in Males
gene: RGS8 /// SDHAP3 | 234297_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0259
p=0.845
p=0.0239
p=0.822
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS8 /// SDHAP3 | 234297_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5601
p=0.0693
p=0.5763
p=0.4997
SmallAirway2
LargeAirway2
Trachea
Macrophage2

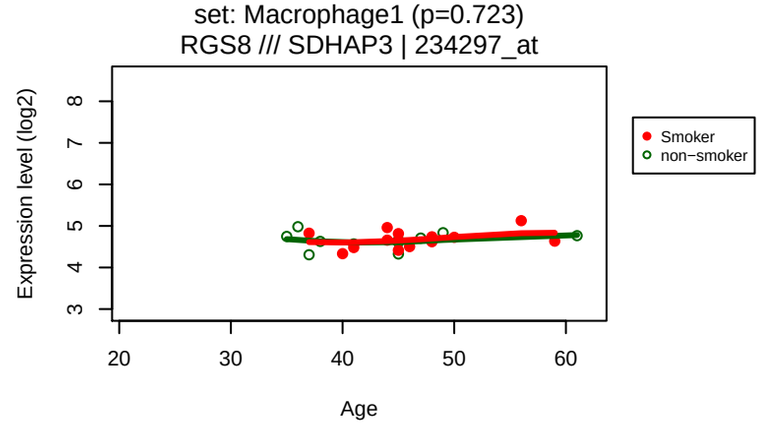

set: SmallAirway1 (p=0.003)
RGS8 /// SDHAP3 | 234297_at
Expression level (log2)
Age
Smoker
non−smoker

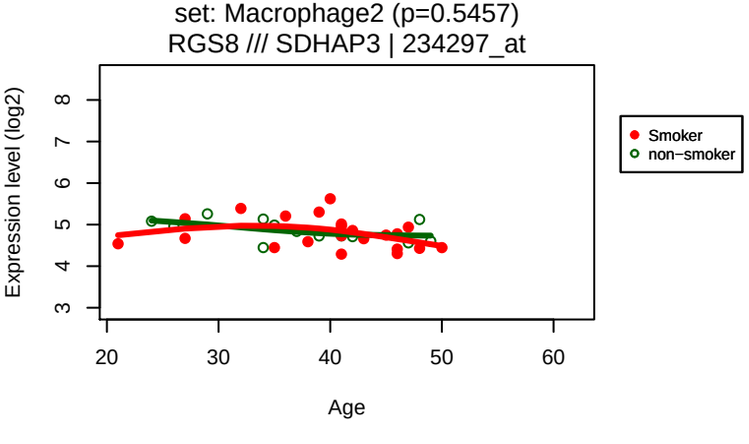

set: SmallAirway2 (p=0.0218)
RGS8 /// SDHAP3 | 234297_at
Expression level (log2)
Age
Smoker
non−smoker

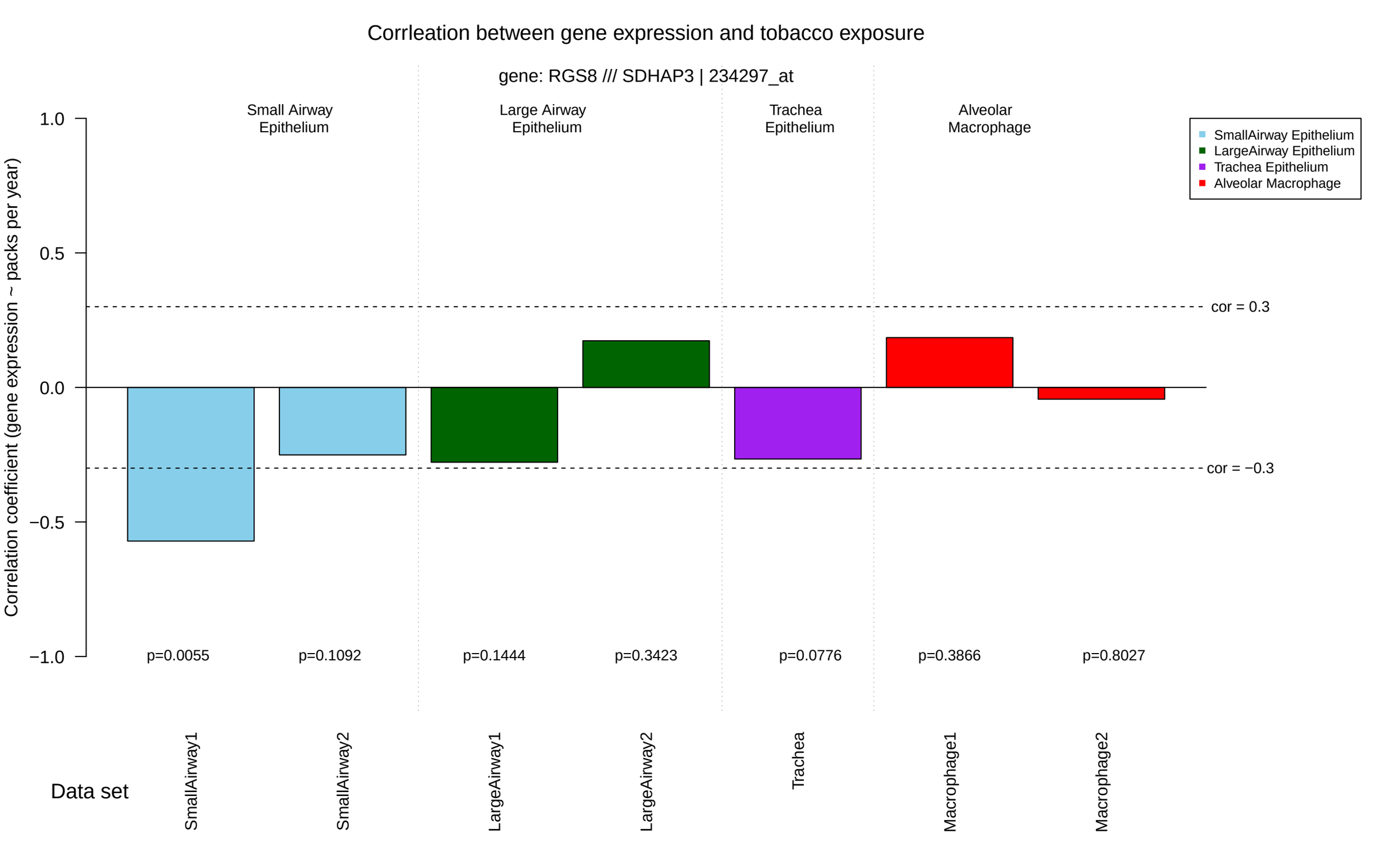

set: LargeAirway1 (p=0.0679)
RGS8 /// SDHAP3 | 234297_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.523)
RGS8 /// SDHAP3 | 234297_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0257)
RGS8 /// SDHAP3 | 234297_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.723)
RGS8 /// SDHAP3 | 234297_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5457)
RGS8 /// SDHAP3 | 234297_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS8 /// SDHAP3 | 234297_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0055
p=0.1092
p=0.1444
p=0.3423
p=0.0776
p=0.3866
p=0.8027
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2